Question 1 [30 marks]

Consider the Digits.Rdata, discussed in class.

(a) Write an R-function to evaluate the digit means and the pooled within-digit covariance. Evaluate the pooled within-digit covariance matrix of the digits data. [**6**]

(b) Carry out a PCA analysis of the covariance from (a). Create a plot showing both the fraction of variance explained by each eigenvector and the cumulative amount of variance explained as a function of the number of eigenvectors retained.[**6**]

Report the minimal number (K) of eigenvectors needed to explain 99% of the overall variance in the data.[**4**]

(c) Using the $784 \times K$ matrix of eigenvectors from (b), evaluate the principal component projection of each digit image onto the set of principal component scores. Put this these scores into an array Z of dimension $K \times 1000 \times 10$ - similar in structure to Digits.Rdata array. [**8**]

Within each digit, evaluate the variance explained by the K eigenvectors identified in (a). Provide a plot showing the percent of variance explained as a function of the digit labels 0-9. [**6**]

Question 2 [30 marks]

Consider the Digits.Rdata, discussed in class.

(a) Use the function *hclust()* with a Euclidean distance matrix and the 'ward.D' clustering method, to reduce the data array for a given digit, d, to a set of $J = 15$ clusters. Evaluate the 784×15 array, C_d, of associated cluster means. [**10**]

(b) For each of the 10000 images, in the data, use lm() to determine the residual sum of squares deviation (rss) between the image and its least squares approximation using each of the 10 C_d arrays. Repeat this for all digits, d, in order to produce a matrix of dimension 10000×10 of rss-values.[**6**]. Provide an array of boxplots, similar to digits2.pdf, with the rss-values by digit.[**6**]

(c) Use the rss array to evaluate the digit, say d_{opt}, that gives the best fit to each digital image in the data.[**4**] Evaluate the re-substitution misclassification percentage rate by digit and add it to the boxplots.[**4**]

Question 3 [30 marks]

The data in table2.dat relates to a collection of 781602 binary star objects observed with the Schmidt telescope at the Palomar Observatory over a 2-year period. Astrophysists have classified each of these star objects into one of 11 types. Analysis of the observed temporal information from the objects provides a set of summary variables in the table. README.pdf is a description file for the data.

Consider the 22 variables [columns 3-24], corresponding to variables labeled 'RAdeg' to 'log(FAP-r)', in the README.pdf file description for table2.dat. The classification of these data into one of 11 types, is provided in the 'Type' variable [column 25].

(a) For each variable evaluate the deviations of values from the variable median and scale this by the variable median absolute deviation. For each object (case) evaluate the average of the absolute deviations across individual variables. Create a dataset in which only the objects with an average absolute deviation below the 99'th percentile of absolute deviations is retained. Report the number of cases removed from each class.[**5**]

(b) (i) Conduct a linear discriminant analysis on the data created in part (a). Use a matplot to show the loading vectors associated with the best 4 discriminant variables and another plot showing the set of 10 F-tests values for assessment of the separation achieved by each of the discriminant variables. Create 10 plots giving labeled boxplots of the 10 discriminant variables obtained. Assemble these 12 plots into a figure (4x3 layout) like that shown in Q3a.pdf but with a suitable legend.[**10**]

(ii) Create a further set of 9 pairwise plots of linear discriminant variables with the mean of each of the 11 object types identified. Assemble these 9 plots into a figure (3x3 layout) like that shown in Q3b.bmp but with top 20 outliers removed and a suitable legend. [**5**]

(c) Using lda() and qda() in the MASS library, carry out linear and quadratic multiple classification analysis of objects using the 22 variables.

(i) For lda() and qda(), report well-formatted tables of re-substitution and cross-validated misclassification rates by class.[**5**]

(ii) Report the overall re-substitution and cross-validated misclassification rates.[**3**]

Comment on the misclassification characteristics obtained - by type and overall.[**2**].